

Johannes Kepler was a German astronomer and mathematician who formulated three laws of planetary motion from observed astronomical data. In providing evidence to support a heliocentric (Sun-centered) Universe, he laid the foundations of modern astronomy and also advanced the study of optics.

A gifted mathematician, in 1589, Johannes Kepler earned a scholarship to the University of Tübingen, where he was tutored by the leading German astronomer Michael Mästlin. Although the university taught geocentric astronomy (with planets revolving around the Earth), Mästlin introduced Kepler to the controversial idea of a heliocentric Universe, first proposed in 1543 by Nicolaus Copernicus (see pp.46–49).

Astronomy and astrology

In 1594, Kepler began teaching mathematics in Graz, Austria, but his spare time was devoted to studying the planets. During the late 1500s, astronomy was of less interest to the wider public than the astrological art of casting horoscopes based on predicted planetary movements.

Astronomers such as Kepler used astrology as their chief source of income and influence, and therefore the accuracy of their predictions was of great significance to their livelihoods.

In 1600, Kepler was invited by Tycho Brahe, a Danish astronomer (see p.58), to be his assistant. Brahe made the most accurate observations of

Kepler made a model of the Universe using five polyhedra (three-dimensional objects with polygon faces). Each was set within a sphere and arranged in concentric sizes.

MILESTONES

HELIOCENTRIC SUPPORT

Produces a written defense of the Copernican model of the Universe in 1596, risking displeasure of the Church.

INFLUENTIAL POSITION

Becomes assistant to Tycho Brahe in 1600; appointed Imperial Mathematician to Rudolph II a year later.

OPTICAL WORK

Begins study of optical theory in 1603; publishes *Astronomiae Pars Optica*, his landmark text, in 1604.

PLANETARY LAWS

Proposes his first two laws of planetary motion in 1609; works for 10 years before publishing his third law.



“Geometry is **one and eternal**, a reflection of the **mind** of God.”

Johannes Kepler, 1610